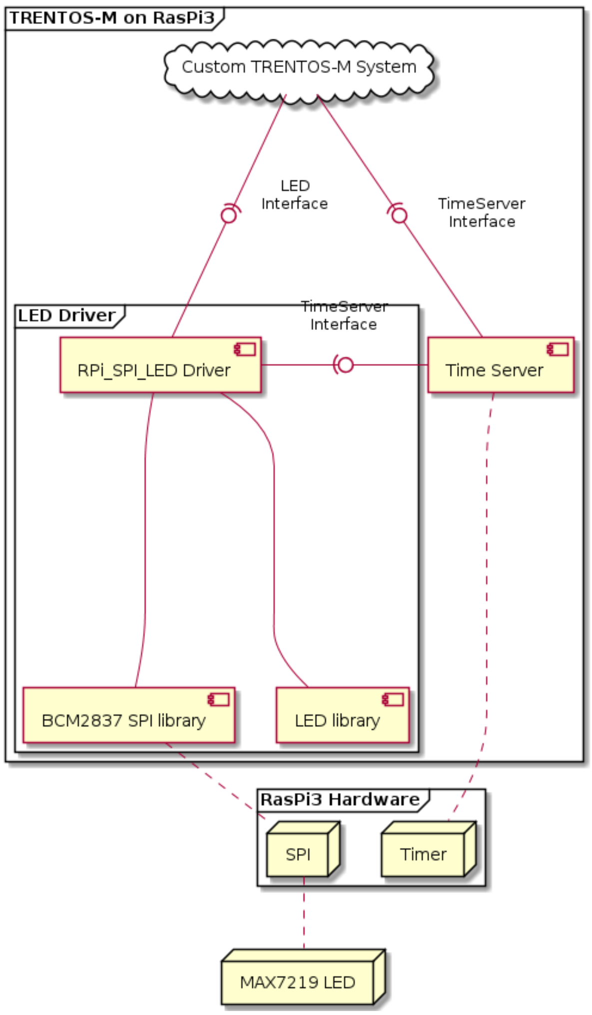
TRENTOS-M SPI LED Demo

General

This demo application scrolls a given string across an 8x8 dot-matrix LED display over SPI.

CAmkES Component Architecture



LED Demo for Raspberry Pi - CAmkES Architecture

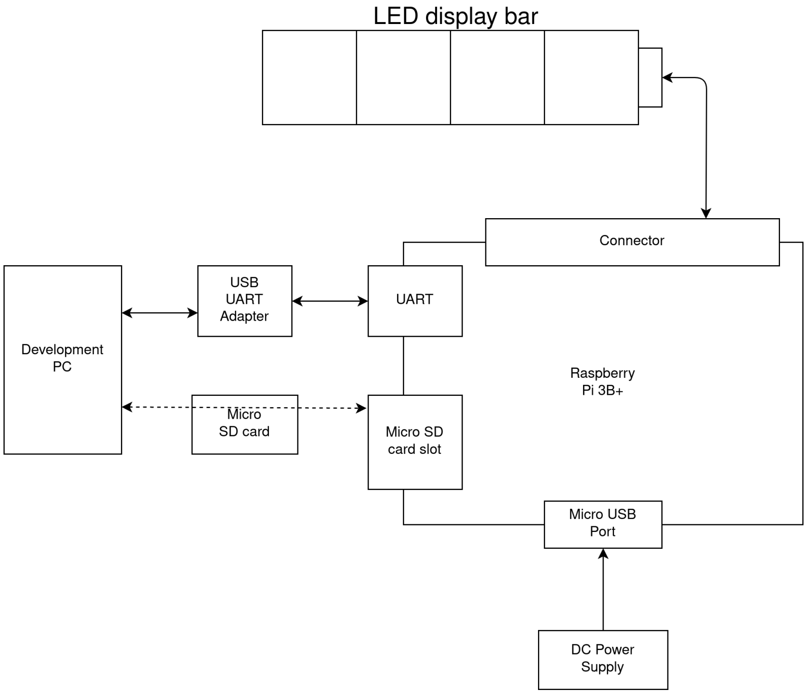
Hardware Setup

This demo is supposed to run standalone on the Raspberry Pi 3 Model B+.

In order to run the demo, the following is required:

- Raspberry Pi 3 Model B+ incl. power supply
- UART-to-USB adapter (see RPi boot process, section "Boot up" for correct wiring) in order to observe RPi behaviour
- MicroSD Card
- 4x8x8 LED display bar

The following diagram shows the hardware setup:



LED Demo for Raspberry Pi - HW Setup
TODO: make picture of setup and upload it
LED Demo for Raspberry Pi - Example Setup

Building the demo

For building the LED demo, the **`build-system.sh`** script has to be used and executed within the **trentos_build** docker container. The following command will invoke this custom build script from inside the build docker container. The container will bind the current working folder to a volume mounted under "/host", execute the script and then self remove.

LED Demo for Raspberry Pi - Call build.sh in the container environment

```
sdk/scripts/open_trentos_build_env.sh sdk/build-system.sh sdk/demos/demo_led_app_rpi3/src rpi3 build-rpi3-Debug-demo_led_app -DCMAKE_BUILD_TYPE=Debug
```

As a result, the folder **build-rpi3-Debug-demo_led_app_rpi3** is created, containing all the build artifacts. The TRENTOS-M system image can be found in **build-rpi3-Debug-demo_led_app_rpi3/images/os_image.bin**.

Please also refer to the "Getting Started" guide for more helpful information on how to build TRENTOS-M systems.

Preparing the demo

The image is flashed on the SD card either through an SD card reader on the laptop or with an SD card switcher.

Prerequisites

This demo assumes that you have the Raspberry Pi required bootup files already installed on the SD Card. If not, make sure to follow the handbook on how to do this. In short, do this:

Copy bootfiles for Raspberry Pi on SD Card

```
cp <path_to_trentos_dir>/sdk/resources/rpi3_sd_card/* <SD_Card_Mount_Dir>/
```

You should also have this SD Card partitioned as *FAT32*. You can use *gparted* in case you still need to partition the SD Card.

Using SD Card reader

Insert the SD Card into the SD card reader.

Get the <block_device_number> of SD card:

Get <block_device_number> of SD Card

```
lsblk
```

Write *os_image.bin* to SD Card:

Write os_image.bin to SD Card

```
sudo mount /dev/<block_device_name> /mnt
cd <path_to_trentos_dir>/build-rpi3-Debug-demo_led_app/images
sudo cp os_image.bin /mnt
sync
sudo umount /mnt
```

Using SD Card switcher

Get <sd_device_number> with:

SD Card switcher - Get SD device number

```
sudo sd-mux-ctrl -l
```

Get <block_device_number> of SD card:

Get block device number of inserted SD card

```
lsblk
```

Copy *os_image.bin* to Raspberry Pi with SD-card switcher:

copy os_image.bin to SD card with SD card switcher

```
sudo sd-mux-ctrl -e <sd_device_number> -s
sudo mount /dev/<block_device_name> /mnt
cd <path_to_trentos_dir>/build-rpi3-Debug-demo_led_app/images
sudo cp os_image.bin /mnt
sync
sudo umount /mnt
sudo sd-mux-ctrl -e <sd_device_number> -d
```

Running the demo

Just boot up the Raspberry Pi and after an initial bootup sequence, the specific text should be scrolled across the LED display.